Evolutionary Biology - Structural Correspondence with the Five-Stage Model

2026-03-21

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Evolutionary Biology

Structural correspondence survey with the five-stage model (Field / Wave / Edge / Vortex / Bundle)

Survey Overview

- **Survey scope**: 10 major theories in evolutionary biology
- **Research question**: Do the various theories of evolutionary biology correspond structurally to the five-stage model?
- **Findings**: Conditional correspondence: 1 case

Overview of the Five-Stage Model

Stage	Definition
Field (ba)	Undifferentiated state. Initial conditions with no direction or structure yet determined
Wave (nami)	Exploratory stage where multiple directions diverge and compete
Edge (en)	Tension state where opposing elements coexist without converging to either side. A place where boundaries are touched, mutual influence occurs, and relationships emerge
Vortex (uzu)	Stage where new coherence (order) spontaneously arises from within tension
Bundle (taba)	Stage where form is fixed and stabilized as a reusable structure

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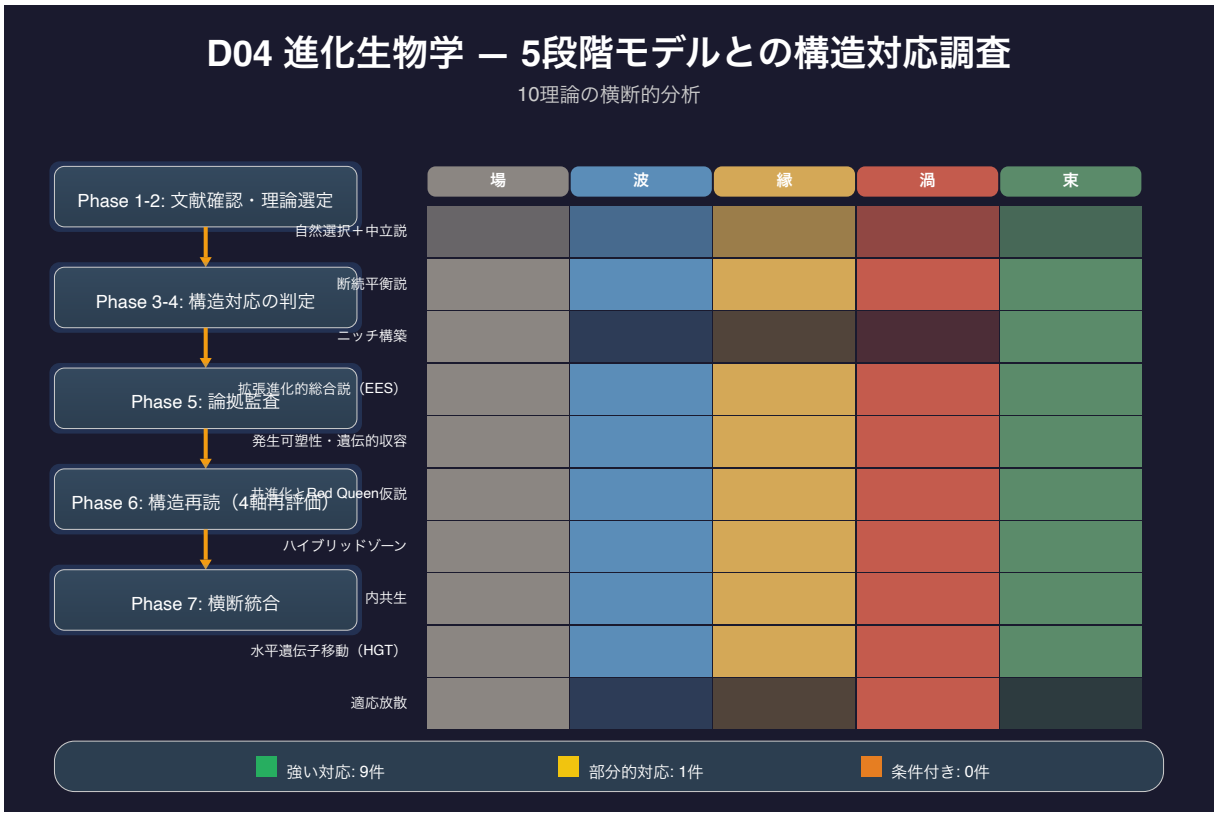

Figure 1: Domain — Research Overview Infographic

Overall Picture of Structural Correspondences

Temperature	Theory Group	Positioning
Near-certain	Natural selection + neutral theory, Developmental plasticity & genetic accommodation, Hybrid zones, Endosymbiosis	Concrete correspondences confirmed across all five stages
Probable	Punctuated equilibrium, Extended Evolutionary Synthesis (EES), Coevolution and Red Queen hypothesis, Horizontal gene transfer	Correspondences exist, but the focus of discussion is on specific stages
Conditional	Niche construction, Adaptive radiation	Correspondences are confirmed, but the mapping concentrates on specific structural patterns (cyclicity or creation from vacancy) rather than the generative process across all five stages

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Major Entry 1: Natural Selection and Neutral Theory (Darwin / Fisher / Kimura / Ohta)

- Natural selection is a mechanism that operates when three conditions—variation, inheritance, and fitness differences—are met. Darwin proposed it in 1859, and it was integrated with Mendelian genetics in the Modern Synthesis of the 1930s–50s. Motoo Kimura's neutral theory (1968) holds

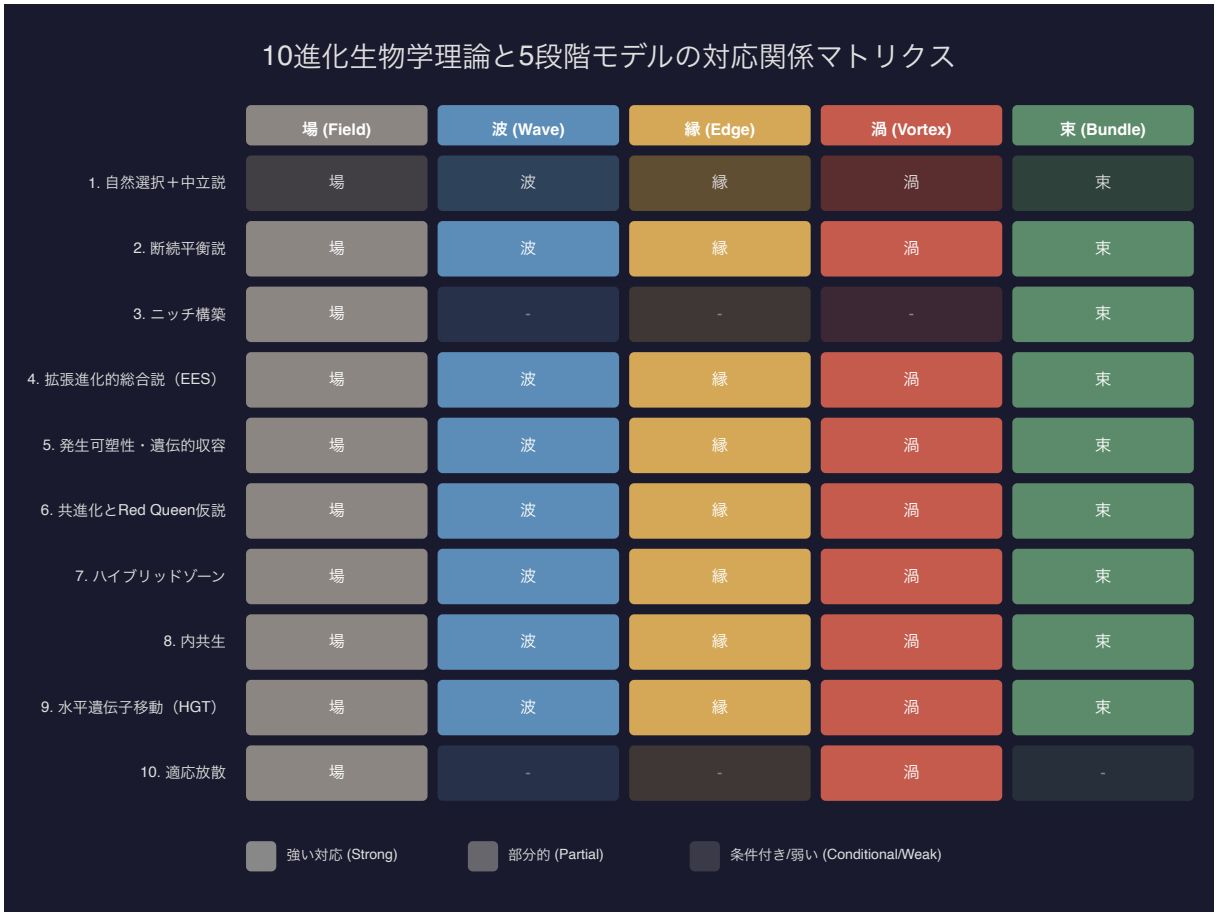

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

that most evolutionary changes at the molecular level are due to genetic drift rather than natural selection. Tomoko Ohta's nearly neutral theory (1973) refined this further, showing that mutations with weak selective effects (|N s| roughly 1) behave neutrally depending on population size.

- **As fact**: Mutations arise during processes such as DNA replication, repair, and recombination. Most new mutations are neutral to mildly deleterious; strongly deleterious mutations are removed by purifying selection. Beneficial mutations are relatively rare but may in some conditions be fixed in a population through selection. Gene duplication can provide raw material for the evolution of new functions, but most copies become inactivated or deleted. Even when preserved, multiple mechanisms have been proposed—not only neofunctionalization but also subfunctionalization and others.
- **As reading**: Here we read a stepwise process from the supply of variation through selection and drift to fixation. The level of analogy is process. Particular attention is given to the sequential structure in which a mutation is retained rather than discarded, and later becomes raw material for a new function. The pattern in which neutral gene duplication subsequently undergoes neofunctionalization demonstrates at the molecular level the structure of "a mutation that was not discarded becomes the raw material for creation."
- **As interpretation**: The pool of genetic and phenotypic diversity corresponds to Field. The phase in which fitness differences become manifest through environmental pressure corresponds to Wave. In this theory, the correspondence of Edge is background in nature—individual-level interactions such as mating, competition, and symbiosis serve that role. The expansion of lineages bearing adaptive traits corresponds to Vortex, and the stabilization as a trait distribution or species characteristic to Bundle. This theory shows the foundational structure of the five stages—the flow from supply of variation (Field) to fixation (Bundle)—with the largest body of data.

Major Entry 2: Punctuated Equilibrium (Eldredge & Gould)

- A theory proposed by Eldredge and Gould in 1972. Drawing on the fossil record, they argued that morphological evolution proceeds through a combination of long periods of stasis and brief periods of rapid change. What is important is the epistemological reversal they proposed against gradualism—the prevailing view that evolution proceeds slowly and continuously: "the discontinuities in the fossil record accurately reflect the reality of evolution rather than being artefacts of an incomplete record."
- **As fact**: In the fossil record, long morphological stasis (on the scale of 100,000–10,000,000 years) combined with brief rapid change (on the scale of 1,000–100,000 years) has been documented in many lineages. Stasis is maintained by stabilizing selection or undirected variation under developmental constraints. Rapid change is explained as allopatric speciation in peripheral isolated populations. However, whether this pattern is universal across all lineages remains debated, and since gradual patterns also exist, some views treat the distribution as a "spectrum."
- **As reading**: Here we read the rhythmic structure of stability and sudden change, and the spatial pattern of "new things arising at the periphery rather than the center." The level of analogy is process, with particular attention to the temporal rhythm of stasis → rapid change and the spatial peripherality of the isolated peripheral population.
- **As interpretation**: The base population during a period of morphological stasis corresponds to Field. The phase in which change pressure rises through environmental fluctuation or geographic isolation corresponds to Wave. The geographic and genetic boundary between the peripheral isolated population and the source population—the place where speciation occurs—corresponds to Edge. The rapid establishment of a new morphology (geologically a "moment" of 1,000–100,000 years) corresponds to Vortex, and stabilization as a new species to Bundle. The structural feature of this theory is that Bundle returns directly to the Field of the next period of stasis.

Major Entry 3: Developmental Plasticity and Genetic Accommodation (West-Eberhard / Waddington)

- Developmental plasticity is the property of a single genotype to produce multiple phenotypes depending on environmental conditions. West-Eberhard (2003) systematized it, and Waddington's

genetic assimilation experiments (1953) are known as a precursor. Genetic assimilation is the phenomenon in which a trait induced by the environment becomes genetically stabilized through selection. Genetic accommodation is a broader concept.

- **As fact**: In phenotypic plasticity, different phenotypes are produced from the same genotype through gene-environment interaction (G×E). In Waddington's 1953 experiment, a phenotype induced by heat shock came to be expressed without heat shock after repeated rounds of selection. Furthermore, research on deep homology demonstrates that diverse structures can be generated independently from a shared set of developmental genes such as Pax6 and Hox.
- **As reading**: Here we read a threshold structure in which the environment switches phenotypes as a boundary condition for development. The level of analogy is mechanism. Particular attention is given to the interface of G×E interaction—the genotype-environment boundary—and to the structure by which this interface biases "which mutations become visible" from the environmental side.
- **As interpretation**: The possibility space of the developmental system—the repertoire of plastic responses—corresponds to Field. Deep homology is the deep structure of this field. The induction of plastic responses by environmental stimuli corresponds to Wave; the G×E interaction interface—where the environment becomes a boundary condition for development and a different phenotype is expressed when a threshold is crossed—to Edge. The expression of a novel phenotype and its spread through the population corresponds to Vortex, and fixation through genetic accommodation to Bundle. The staged progression from plasticity to fixation is a structure that corresponds directly to the five stages.

Cross-Cutting Patterns

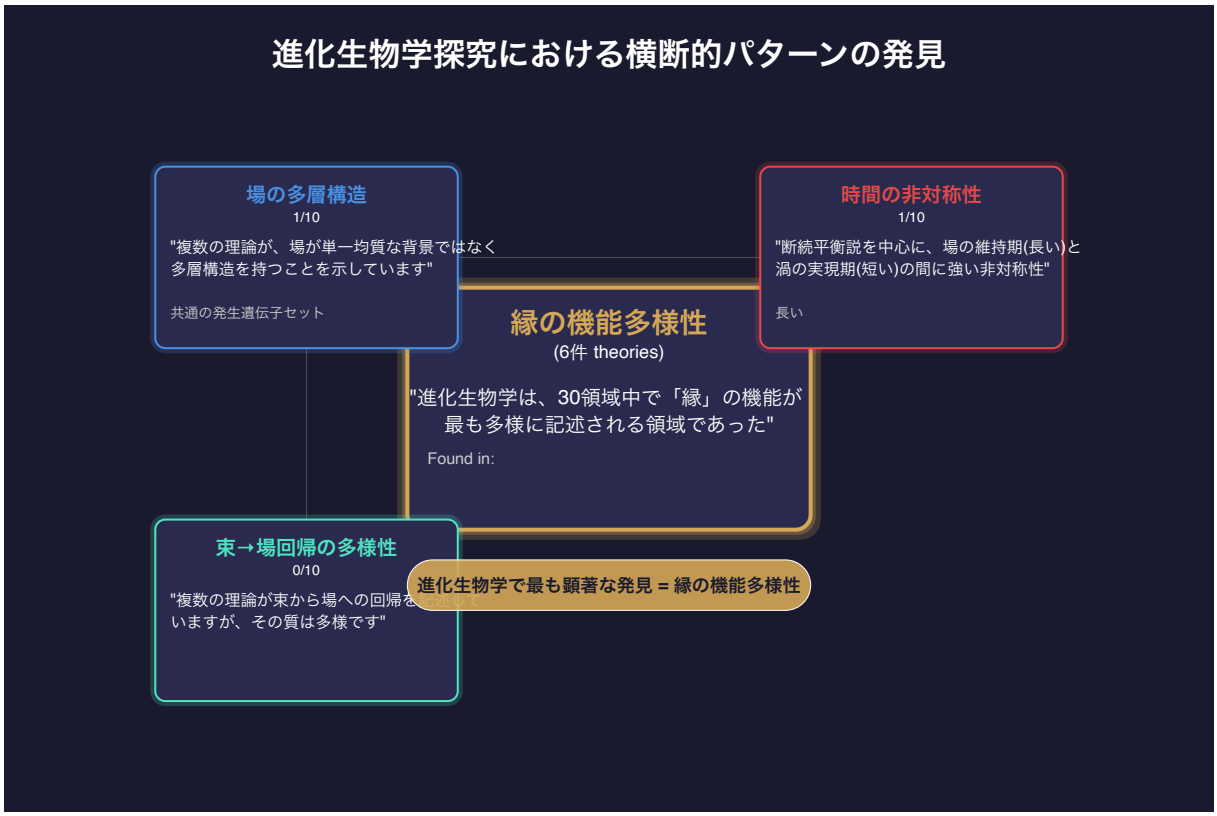

Figure 3: Domain — Cross-Cutting Patterns Diagram

- The most conspicuous cross-cutting pattern in evolutionary biology is the functional diversity of "Edge"
- **Maintained Edge**: In hybrid zones, the boundary persists stably while retaining thickness measured by cline width

- **Driving Edge**: In the Red Queen hypothesis, the inter-species boundary functions as a dynamic device that continuously generates selective pressure
- **Fusing Edge**: In endosymbiosis, a new hierarchy emerges through the disappearance of the boundary (cell membrane)

Unresolved Questions

- Whether the four patterns of "Edge" (maintained, driving, fusing, generating) are diversity specific to evolutionary biology or are equally confirmed in other domains requires cross-domain verification.
- How to reflect in the five-stage model the finding that Field has "types"—the distinction between communal field and individuated field—is unsettled. If Field has qualitative diversity beyond mere "undifferentiated initial conditions," it becomes necessary to expand the first half of the model.
- How to handle the pattern shown by the Red Queen hypothesis—"Bundle does not stabilize"—is an open question. A decision is needed as to whether to redefine Bundle as "provisional stability" or to treat the cycle that does not reach Bundle as a separate category.
- When discussing structural correspondences in evolutionary biology, the methodological tension between the basic fact that natural selection has no intention and describing each phase of the five-stage model as a "process" is something that should always be kept in mind.

Conclusion

- This survey confirmed that evolutionary biology is a domain with overall strong structural similarity to the five-stage model
- Near-certain are: natural selection + neutral theory (the foundational structure of variation → selection → fixation), developmental plasticity and genetic accommodation (the threshold structure of G×E and staged fixation), hybrid zones (quantification of Edge and two possible outcomes), and endosymbiosis (hierarchical emergence through boundary disappearance)
- The greatest insight gained from evolutionary biology is that "Edge" exhibits not a single function but at least four patterns, and that the circular structure in which "Bundle" rewrites the preconditions of the next "Field" lies at the core of the theories